Shocks, Weak Solutions, and Entropy: An Exposition of Burgers' Equation

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1 Motivation

1.1 Origin in fluid dynamics

Burgers' equation was introduced by J. M. Burgers as a simplified model for turbulence [1], isolating the two competing effects that make the Navier–Stokes equations hard to analyse: nonlinear advection, which steepens a velocity profile, and viscous diffusion, which smooths it. The one-dimensional Navier–Stokes momentum equation is

$$u_t + u\,u_x = -\frac{1}{\rho}p_x + \nu\,u_{xx}, \tag{1}$$

where $u(x,t)$ is velocity, ρ density, p pressure, and $\nu > 0$ viscosity. In more than one dimension, p is determined implicitly by the incompressibility constraint $\nabla \cdot u = 0$; this nonlocal coupling is what makes Navier–Stokes hard to control. In one dimension, $u_x = 0$ would force u constant, so the constraint is dropped along with the pressure term entirely, leaving

$$u_t + u\,u_x = \nu\,u_{xx}, \tag{2}$$

the *viscous Burgers' equation*. Setting $\nu = 0$ gives the *inviscid* equation

$$u_t + u\,u_x = 0, \tag{3}$$

which is the main object of study in Sections 2–7, before viscosity is reintroduced in Section 8. What remains after this simplification is a single scalar equation depending only on local values of u, but one that retains the nonlinear term $u\,u_x$ responsible for the spontaneous formation of discontinuities studied from Section 3 onward.

Remark 1.1 (Vorticity-free flow). Without the constraint $\nabla \cdot u = 0$, the flow may be taken irrotational, so $u = \varphi_x$ for a scalar potential φ. This is what allows the Cole–Hopf transformation (Section 8) to linearise (2) into the heat equation.

1.2 Burgers' equation as a scalar conservation law

Equations (3) and (2) are instances of a *scalar conservation law*,

$$u_t + f(u)_x = 0, \tag{4}$$

with flux f, together with its viscous regularisation $u_t + f(u)_x = \nu\,u_{xx}$. For Burgers' equation,

$$f(u) = \frac{u^2}{2}, \tag{5}$$

so $f(u)_x = u\,u_x$ recovers (3). Such equations describe a locally conserved quantity transported at state-dependent speed $f'(u)$, and arise throughout gas dynamics and multiphase flow. Burgers' equation is the simplest nontrivial case (a linear f gives only the trivial transport equation $u_t + c\,u_x = 0$), and is the standard first example for the general theory of characteristics, shocks, weak solutions, and entropy conditions developed in Sections 2–6.

1.3 Why study Burgers' equation

Three features make Burgers' equation worth studying in its own right. First, it is the simplest equation whose smooth solutions can lose smoothness in finite time despite smooth data and no forcing; because it is scalar and one-dimensional, this can be analysed completely, unlike the analogous (and in three dimensions still open) questions for Euler and Navier–Stokes. Second, once weak solutions are admitted, it exhibits genuine non-uniqueness (Section 5), requiring an entropy condition (Section 6) to restore well-posedness — a phenomenon common to conservation laws in general, studied here with minimal technical overhead. Third, the viscous equation (2) is one of the few nonlinear PDEs solvable in closed form, via the Cole–Hopf transformation (Section 8), allowing a direct, rigorous comparison between the viscous and inviscid theories.

2 Method of Characteristics

This section derives an implicit solution formula for the inviscid Burgers' equation via the method of characteristics. The formula's eventual breakdown is what forces the more general framework of Sections 3–6.

2.1 Setup

Consider the initial value problem

$$\begin{cases} u_t + u\,u_x = 0, & x \in \mathbb{R},\ t > 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \tag{6}$$

with $u_0 \in C^1(\mathbb{R})$. The method of characteristics looks for curves $x(t)$ along which the PDE reduces to an ODE for $u(x(t),t)$.

2.2 Derivation of the characteristic equations

For a solution u of (6) and a curve $x(t)$, the chain rule gives

$$\frac{d}{dt}\,u(x(t),t) = u_x(x(t),t)\,\dot{x}(t) + u_t(x(t),t). \tag{7}$$

Since $u_t = -u\,u_x$ on solutions, choosing $x(t)$ to satisfy

$$\dot{x}(t) = u(x(t),t) \tag{8}$$

reduces (7) to

$$\frac{d}{dt}\,u(x(t),t) = 0. \tag{9}$$

Curves satisfying (8) are *characteristics*, and u is constant along each one.

2.3 Characteristics are straight lines

Proposition 2.1 (Characteristics are straight lines)**.** *Let u be a C^1 solution of (6) on $\mathbb{R}\times[0,T)$, and let $x(t)$ solve (8) with $x(0)=x_0$. Then for $0 \le t < T$,*

$$u(x(t),t) = u_0(x_0), \qquad x(t) = x_0 + u_0(x_0)\,t. \tag{10}$$

Proof. By (9), $u(x(t),t) \equiv u(x_0,0) = u_0(x_0)$. Substituting this constant into (8) gives $\dot{x}(t) = u_0(x_0)$, which integrates to the stated formula for $x(t)$. □

Thus any classical solution is completely determined by (10), reducing its construction to inverting $x_0 \mapsto x = x_0 + u_0(x_0)\,t$ at each fixed t.

2.4 The implicit solution formula

Theorem 2.2 (Implicit solution via characteristics)**.** *Let $u_0 \in C^1(\mathbb{R})$. For each $t \geq 0$ such that $\Phi_t : \mathbb{R} \to \mathbb{R}$, $\Phi_t(x_0) = x_0 + u_0(x_0)\,t$, is a bijection with C^1 inverse, the function u defined implicitly by*

$$u(x,t) = u_0(x_0), \qquad \text{where } x = x_0 + u_0(x_0)\,t, \tag{11}$$

is the unique C^1 solution of (6) at time t.

Proof. Well-definedness. Bijectivity of Φ_t means $u(x,t) = u_0(\Phi_t^{-1}(x))$ is well defined, and C^1 since Φ_t^{-1} and u_0 are.

The PDE holds. Write $x_0 = x_0(x,t) = \Phi_t^{-1}(x)$. Differentiating $x = x_0 + u_0(x_0)\,t$ in x and in t gives

$$\frac{\partial x_0}{\partial x} = \frac{1}{1 + u_0'(x_0)\,t}, \qquad \frac{\partial x_0}{\partial t} = \frac{-u_0(x_0)}{1 + u_0'(x_0)\,t}, \tag{12}$$

both well defined since $\Phi_t'(x_0) = 1 + u_0'(x_0)t \neq 0$. By the chain rule applied to $u(x,t) = u_0(x_0(x,t))$,

$$u_x = \frac{u_0'(x_0)}{1 + u_0'(x_0)\,t}, \qquad u_t = \frac{-u_0(x_0)\,u_0'(x_0)}{1 + u_0'(x_0)\,t}, \tag{13}$$

so $u_t + u\,u_x = 0$, and $u(x,0) = u_0(x)$ since $\Phi_0 = \mathrm{id}$.

Uniqueness. Any C^1 solution must, by Proposition 2.1, agree with (11) along every characteristic. □

Corollary 2.3 (Local existence)**.** *For $u_0 \in C^1(\mathbb{R})$ with u_0' bounded below, there exists $T > 0$ such that (6) has a unique C^1 solution on $\mathbb{R} \times [0,T)$, given by (11).*

Proof sketch. It suffices that $\Phi_t'(x_0) = 1 + u_0'(x_0)t$ stay bounded away from 0 uniformly in x_0 for $t \in [0,T)$; monotonicity of Φ_t then gives bijectivity with C^1 inverse. Let $m = \inf_{x_0} u_0'(x_0)$. If $m \geq 0$, then $\Phi_t'(x_0) \geq 1$ for all $t \geq 0$ and $T = \infty$ works. If $m < 0$, then $\Phi_t'(x_0) \geq 1 + mt > 0$ for $t < -1/m$, so any $T \leq -1/m$ works. □

The bound $T \leq -1/m$ is not an artefact of the proof: it is the exact breakdown time, as Proposition 3.1 shows.

2.5 Worked example

Let $u_0(x) = -\arctan(x)$, so $u_0'(x) = -1/(1+x^2) \in [-1, 0)$. By Theorem 2.2, the solution satisfies

$$u(x,t) = -\arctan(x_0), \qquad x = x_0 - \arctan(x_0)\,t, \tag{14}$$

valid while $\Phi_t'(x_0) = 1 - t/(1+x_0^2) > 0$ for all x_0. Since $1/(1+x_0^2)$ is maximised at $x_0 = 0$, this holds exactly for $t < 1$: the characteristic through the origin, carrying the steepest slope $u_0'(0) = -1$, is the first to cause Φ_t to fail to be injective, at $t = 1$.

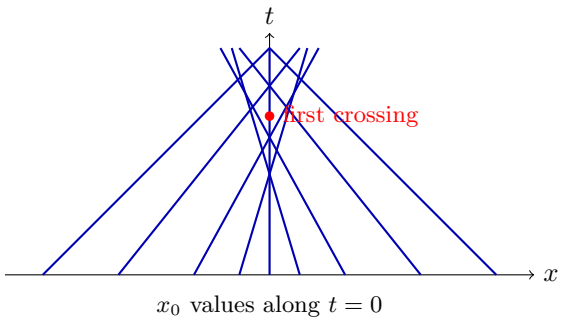


Figure 1: Characteristics for decreasing initial data u_0. Where u_0 is decreasing, characteristics converge; the first crossing time is $T = -1/\min u_0'$ (Corollary 2.3), computed precisely in Section 3.

Remark 2.4 (What happens beyond T). Once two characteristics carrying different values of u_0 meet, formula (11) assigns u two values at once, so no single-valued classical solution can exist beyond this time. Making sense of the equation past this point is the subject of Sections 3 and 4.

3 Shock Formation

This section computes the exact time and location at which the classical solution of Section 2 first breaks down.

3.1 Blow-up of the derivative

From the proof of Theorem 2.2,

$$u_x(x,t) = \frac{u_0'(x_0)}{1 + u_0'(x_0)\,t}, \qquad x = x_0 + u_0(x_0)\,t. \tag{15}$$

If $u_0'(x_0) \geq 0$, the denominator stays ≥ 1 and u_x remains bounded along that characteristic for all $t \geq 0$. If $u_0'(x_0) < 0$, the denominator vanishes at the finite time $t = -1/u_0'(x_0)$, where (15) shows $|u_x| \to \infty$: the solution value stays bounded, but its slope becomes vertical.

3.2 The envelope of the characteristics

The characteristics form a one-parameter family of lines

$$x(x_0,t) = x_0 + u_0(x_0)\,t, \qquad x_0 \in \mathbb{R}. \tag{16}$$

The first global breakdown occurs where neighbouring characteristics meet soonest, i.e. on the envelope of this family, found by solving (16) together with $\partial x/\partial x_0 = 0$:

$$1 + u_0'(x_0)\,t = 0. \tag{17}$$

This is exactly the vanishing of $\Phi_t'(x_0)$, so the envelope condition coincides with the derivative blow-up condition above.

3.3 The shock formation time

Proposition 3.1 (Shock formation time)**.** *Let $u_0 \in C^1(\mathbb{R})$ with $m := \min_{x_0} u_0'(x_0) < 0$, attained at $x_0 = a$. Then the classical solution exists and is C^1 on $\mathbb{R} \times [0, T^*)$ for*

$$T^* = \frac{-1}{m}, \tag{18}$$

and fails to extend as a single-valued C^1 function beyond T^*. The first breakdown occurs at

$$x^* = a + u_0(a)\,T^*, \qquad t^* = T^*. \tag{19}$$

Proof. Since $s \mapsto -1/s$ is strictly increasing on $(-\infty, 0)$ and $u_0'(x_0) \geq m$ everywhere, $-1/u_0'(x_0) \geq -1/m = T^*$ for every x_0 with $u_0'(x_0) < 0$, with equality only at $x_0 = a$. So T^* is the smallest time at which any characteristic meets a neighbour: $\Phi_t'(x_0) > 0$ for all x_0 throughout $[0, T^*)$, giving a classical solution there by Theorem 2.2, while $\Phi_{T^*}'(a) = 0$ makes Φ_{T^*} fail to be locally invertible at a, so u_x is unbounded near (x^*, t^*) by (15). Substituting $x_0 = a$, $t = T^*$ into (16) gives x^*. □

3.4 Worked example, continued

For $u_0(x) = -\arctan(x)$, $u_0'(x_0) = -1/(1 + x_0^2)$ is minimised at $a = 0$ with $m = -1$, so Proposition 3.1 gives $T^* = 1$, matching Section 2.5. The shock location is

$$x^* = 0 + (-\arctan(0)) \cdot 1 = 0, \tag{20}$$

so the singularity first appears at $(x^*, t^*) = (0, 1)$, matching Figure 1.

3.5 From breakdown to weak solutions

Proposition 3.1 shows the classical formulation of (6) cannot describe decreasing initial data beyond T^*, even though nothing in the underlying physics suggests the evolution should stop. Section 4 resolves this by weakening the notion of solution: rather than requiring $u_t + u\,u_x = 0$ pointwise, u need only satisfy an integral identity against smooth test functions, which admits discontinuous solutions for all $t \geq 0$.

4 Weak Solutions and the Integral Form

Since classical solutions of (6) cannot be extended past the shock time T^* of Proposition 3.1, this section introduces a weaker notion of solution that admits discontinuities, and derives the condition such discontinuities must satisfy.

4.1 The integral formulation

Let u be a C^1 solution of $u_t + f(u)_x = 0$ on $\mathbb{R} \times (0,\infty)$ with $f(u) = u^2/2$, and let $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$ be any test function, compactly supported but not required to vanish at $t = 0$. Multiplying the PDE by φ, integrating over $\mathbb{R} \times (0,\infty)$, and integrating by parts in both x and t (using that φ has compact support, so all boundary terms at spatial infinity vanish) gives

$$0 = \int_0^\infty \int_{\mathbb{R}} \big(u_t + f(u)_x\big)\varphi \, dx\, dt = -\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt - \int_{\mathbb{R}} u_0(x)\,\varphi(x,0)\, dx, \tag{21}$$

where the boundary term at $t = 0$ survives because $\varphi(x,0)$ need not vanish. Equation (21) makes sense whenever $u \in L^\infty$, with no differentiability required of u at all: this motivates the following definition.

Definition 4.1 (Weak solution)**.** A function $u \in L^\infty(\mathbb{R} \times (0,\infty))$ is a *weak solution* of (6) if

$$\int_0^\infty \int_{\mathbb{R}} \big(u\,\varphi_t + f(u)\,\varphi_x\big)\, dx\, dt + \int_{\mathbb{R}} u_0(x)\,\varphi(x,0)\, dx = 0 \tag{22}$$

for every $\varphi \in C_c^1(\mathbb{R} \times [0,\infty))$.

4.2 Consistency with classical solutions

Proposition 4.2 (Consistency)**.** *If $u \in C^1(\mathbb{R} \times [0,\infty))$ satisfies* (6) *classically, it is a weak solution. Conversely, if a weak solution u happens to lie in $C^1(\mathbb{R} \times [0,\infty))$, it satisfies* (6) *classically.*

Proof sketch. The first claim is precisely the computation in (21), read in reverse. For the converse, reversing the integration by parts shows a C^1 weak solution satisfies $\int (u_t + f(u)_x)\varphi + \int (u(x,0) - u_0(x))\varphi(x,0) = 0$ for every $\varphi \in C_c^1$; taking φ supported away from $t = 0$ forces $u_t + f(u)_x = 0$ pointwise by the fundamental lemma of the calculus of variations, and taking $\varphi(\cdot,0)$ arbitrary then forces $u(x,0) = u_0(x)$. □

4.3 The Rankine–Hugoniot condition

The purpose of Definition 4.1 is to admit solutions with jump discontinuities. The next result identifies exactly what constraint such a jump must satisfy.

Theorem 4.3 (Rankine–Hugoniot condition)**.** *Let $s : [0,\infty) \to \mathbb{R}$ be C^1, and suppose u is a weak solution of* (6) *that is C^1 on each side of the curve $x = s(t)$, with one-sided limits*

$$u_\ell(t) = \lim_{x \to s(t)^-} u(x,t), \qquad u_r(t) = \lim_{x \to s(t)^+} u(x,t) \tag{23}$$

existing and $u_\ell(t) \neq u_r(t)$. Then u satisfies $u_t + f(u)_x = 0$ classically away from the curve, and along the curve

$$s'(t) = \frac{f(u_\ell(t)) - f(u_r(t))}{u_\ell(t) - u_r(t)}. \tag{24}$$

Proof. Fix $\varphi \in C_c^1(\mathbb{R} \times (0,\infty))$ supported in a region crossed by the curve $x = s(t)$, and split the domain of integration in (22) into $\Omega_\ell = \{x < s(t)\}$ and $\Omega_r = \{x > s(t)\}$, on each of which u is C^1. On each piece, integrate by parts in the reverse direction to (21), which is valid since u is C^1 there; this produces the interior term $\iint (u_t + f(u)_x)\varphi$ (which vanishes if u solves the PDE classically away from the curve, established by the same interior argument as in Proposition 4.2 applied to test functions supported in Ω_ℓ or Ω_r alone) plus a boundary term along $x = s(t)$ from each side. Since the outward unit normal to Ω_ℓ along the curve is proportional to $(1, -s'(t))$ (and to $(-1, s'(t))$ for Ω_r), these boundary contributions combine to give

$$\int_0^\infty \Big[\big(f(u_\ell(t)) - f(u_r(t))\big) - s'(t)\big(u_\ell(t) - u_r(t)\big)\Big]\varphi(s(t),t)\, dt = 0. \tag{25}$$

Since $\varphi(s(t),t)$ is an arbitrary C^1 function of t along the curve, the bracketed term must vanish identically, which is (24). □

For Burgers' flux $f(u) = u^2/2$, (24) simplifies using $f(u_\ell) - f(u_r) = \frac{1}{2}(u_\ell - u_r)(u_\ell + u_r)$ to the familiar average-speed formula

$$s'(t) = \frac{u_\ell(t) + u_r(t)}{2}. \tag{26}$$

Remark 4.4. Equation (26) already appears implicitly in Section 3: at the moment of first breakdown, the emerging discontinuity moves at the average of the characteristic speeds u_ℓ and u_r arriving from either side, consistent with the characteristic picture of Figure 1. Section 7 uses (26) to construct explicit shock solutions.

5 Non-Uniqueness of Weak Solutions

Definition 4.1 was introduced to admit discontinuous solutions past the shock time of Proposition 3.1. This section shows that it admits *too many*: a single initial condition can have more than one weak solution, so the weak formulation alone does not determine the evolution.

5.1 The Riemann problem

Consider (6) with the specific piecewise-constant data

$$u_0(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases} \tag{27}$$

This is the simplest instance of a *Riemann problem*, and is studied in general in Section 7. Since u_0 is increasing across $x = 0$, the method of characteristics of Section 2 does not directly apply: characteristics emanate from $x_0 < 0$ with speed 0 and from $x_0 > 0$ with speed 1, so no characteristic covers the region $0 < x < t$ at all. Two different weak solutions are constructed below, both consistent with this characteristic picture away from $0 < x < t$.

5.2 A shock solution

Define

$$u_A(x,t) = \begin{cases} 0, & x < st, \\ 1, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2} = \frac{0+1}{2} = \frac{1}{2}, \tag{28}$$

where s is chosen exactly to satisfy the Rankine–Hugoniot condition (26) with $u_\ell = 0$, $u_r = 1$.

Proposition 5.1. *u_A is a weak solution of* (6) *with data* (27).

Proof. Away from the line $x = st$, u_A is locally constant, hence C^1 and trivially satisfies $u_t + f(u)_x = 0$. Along $x = st$, u_A has a jump from $u_\ell = 0$ to $u_r = 1$ moving at exactly the Rankine–Hugoniot speed s. By Theorem 4.3 (read in the direction verified in its proof: a piecewise C^1 function solving the PDE on each side and satisfying (24) across the jump satisfies the weak formulation (22)), u_A is a weak solution. □

5.3 A rarefaction solution

Define instead

$$u_B(x,t) = \begin{cases} 0, & x \le 0, \\ x/t, & 0 < x < t, \\ 1, & x \ge t. \end{cases} \tag{29}$$

Proposition 5.2. *u_B is a weak solution of* (6) *with data* (27).

Proof. On each of the three open regions $x < 0$, $0 < x < t$, and $x > t$, u_B is C^1. On $0 < x < t$, direct differentiation gives $(u_B)_t = -x/t^2$ and $(u_B)_x = 1/t$, so

$$(u_B)_t + u_B(u_B)_x = -\frac{x}{t^2} + \frac{x}{t} \cdot \frac{1}{t} = 0, \tag{30}$$

and the PDE holds classically on each piece. Moreover u_B is *continuous* across both boundary lines $x = 0$ and $x = t$ (both one- sided limits equal 0 at $x = 0$ and 1 at $x = t$), unlike u_A. Repeating the boundary computation in the proof of Theorem 4.3 across each of these lines, the jump terms $f(u_\ell) - f(u_r) - s'(u_\ell - u_r)$ vanish identically because $u_\ell = u_r$ there (there is no jump to begin with, regardless of the line's speed s'). Hence all boundary contributions vanish and u_B satisfies (22). □

5.4 Two solutions, one initial condition

Corollary 5.3 (Non-uniqueness)**.** *The initial value problem* (6) *with data* (27) *has at least two distinct weak solutions,* u_A *and* u_B*.*

This is a genuine failure of uniqueness, not merely a technicality: u_A and u_B differ on the entire wedge $0 < x < t$, and both are constructed from, and consistent with, the same tools developed so far — u_A from the Rankine–Hugoniot condition of Section 4, and u_B from a direct (if unmotivated) guess later justified by the general construction of Section 7. Definition 4.1 alone gives no way to discard either one.

Remark 5.4. Physically, only u_B is correct: for the fluid interpretation of Section 1, increasing data $u_\ell < u_r$ should disperse smoothly (a rarefaction), not jump discontinuously (u_A is, in effect, a shock running *backwards* in time). Section 6 makes this precise with a condition that u_A fails and u_B satisfies, restoring uniqueness.

6 The Entropy Condition

Corollary 5.3 showed that the weak formulation alone does not determine a unique solution. This section introduces an additional condition, motivated by the vanishing-viscosity limit made precise in Section 8, that selects the physically correct weak solution among any competing candidates.

6.1 The Lax entropy condition

Definition 6.1 (Lax entropy condition)**.** Let u be a weak solution of a scalar conservation law $u_t + f(u)_x = 0$ with f convex, having a jump discontinuity along a C^1 curve $x = s(t)$ with left and right limits $u_\ell(t)$, $u_r(t)$. The jump satisfies the *Lax entropy condition* if

$$f'(u_\ell(t)) > s'(t) > f'(u_r(t)) \tag{31}$$

for all t. A weak solution is an *entropy solution* if every jump discontinuity it contains satisfies (31).

For Burgers' equation, $f'(u) = u$, so (31) together with the Rankine–Hugoniot speed $s' = (u_\ell + u_r)/2$ reduces to

$$u_\ell > \frac{u_\ell + u_r}{2} > u_r \quad \Longleftrightarrow \quad u_\ell > u_r. \tag{32}$$

Remark 6.2. Condition (31) says the characteristics on either side of the shock must run *into* it as t increases, rather than out of it: information is being destroyed at the shock (consistent with physical irreversibility), not created. This is the precise sense in which a shock differs from a rarefaction, where characteristics instead fan *out* of a single point (Section 5.3).

6.2 Application to the Riemann problem of Section 5

Proposition 6.3. *For the Riemann data* (27) *(*$u_\ell = 0 < u_r = 1$*), the shock solution* u_A *of* (28) *is not an entropy solution, while the rarefaction solution* u_B *of* (29) *is.*

Proof. For u_A, condition (32) requires $u_\ell > u_r$, but here $u_\ell = 0 < 1 = u_r$, so (31) fails: u_A is not an entropy solution. The function u_B contains no jump discontinuity at all (Proposition 5.2 shows it is continuous), so Definition 6.1 imposes no condition on it, and it is vacuously an entropy solution. □

Proposition 6.3 recovers, as a formal criterion, the physical judgement made informally at the end of Section 5: among the two weak solutions constructed there, the entropy condition selects exactly the rarefaction u_B and rules out the backwards-running shock u_A.

6.3 Uniqueness

The entropy condition is not merely a rule of thumb for discarding individual bad solutions: it in fact restores full well-posedness.

Theorem 6.4 (Oleinik, uniqueness of entropy solutions; see [2], §3.4)**.** *For* $u_0 \in L^\infty(\mathbb{R})$ *and* f *uniformly convex, the initial value problem* $u_t + f(u)_x = 0$*,* $u(\cdot, 0) = u_0$*, has at most one entropy solution satisfying* (31) *at every jump.*

The proof of Theorem 6.4 (via an L^1-contraction argument comparing two entropy solutions) is beyond the scope of this paper and is not reproduced here; it is stated to record that the selection principle of Definition 6.1 is exactly strong enough to recover uniqueness, and no stronger condition is needed.

7 The Riemann Problem

Sections 5–6 treated the specific data $u_\ell = 0 < u_r = 1$. This section gives the complete entropy solution for arbitrary constant states $u_\ell, u_r \in \mathbb{R}$, i.e. the general *Riemann problem*

$$u_0(x) = \begin{cases} u_\ell, & x < 0, \\ u_r, & x > 0. \end{cases} \tag{33}$$

7.1 The rarefaction fan

The solution $u_B(x,t) = x/t$ constructed in Section 5.3 is a special case of a *self-similar* solution, i.e. one of the form $u(x,t) = g(x/t)$. Writing $\xi = x/t$ and substituting into the PDE gives

$$u_t + u\,u_x = -\frac{\xi}{t}g'(\xi) + \frac{g(\xi)}{t}g'(\xi) = \frac{g'(\xi)}{t}\big(g(\xi) - \xi\big) = 0, \tag{34}$$

so any self-similar solution satisfies either $g'(\xi) = 0$ (a constant state) or $g(\xi) = \xi$ (the fan $u = x/t$) on each interval of ξ. This is exactly the structure needed to connect two constant states, provided $u_\ell < u_r$: since characteristic speed equals u, the states u_ℓ and u_r correspond to characteristics of speed u_ℓ and u_r, and the fan $u = x/t$ smoothly interpolates the speeds in between whenever $u_\ell < u_r$.

7.2 The complete solution

Theorem 7.1 (Entropy solution of the Riemann problem)**.** *The unique entropy solution of* (6) *with data* (33) *is:*

(a) *If $u_\ell > u_r$ (shock case),*

$$u(x,t) = \begin{cases} u_\ell, & x < st, \\ u_r, & x > st, \end{cases} \qquad s = \frac{u_\ell + u_r}{2}. \tag{35}$$

(b) *If $u_\ell \le u_r$ (rarefaction case),*

$$u(x,t) = \begin{cases} u_\ell, & x \le u_\ell t, \\ x/t, & u_\ell t < x < u_r t, \\ u_r, & x \ge u_r t. \end{cases} \tag{36}$$

Proof. (a) As in Proposition 5.1, u is locally constant away from $x = st$ and jumps at the Rankine–Hugoniot speed $s = (u_\ell + u_r)/2$ across it, so u is a weak solution by Theorem 4.3. Since $u_\ell > u_r$, the Lax condition (32) holds, so u is an entropy solution; uniqueness is Theorem 6.4.

(b) The case $u_\ell = u_r$ is the trivial constant solution, so suppose $u_\ell < u_r$. On each of the three regions in (36), u is C^1 and solves the PDE classically: on the outer two by inspection, and on the fan $u_\ell t < x < u_r t$ by Section 7.1. At the two interior boundaries $x = u_\ell t$ and $x = u_r t$, u is continuous (both one-sided limits agree, equalling u_ℓ and u_r respectively), so exactly as in Proposition 5.2 the boundary terms in the weak formulation vanish and u is a weak solution. It contains no jump discontinuity, so it is vacuously an entropy solution, and uniqueness is again Theorem 6.4. □

Taking $(u_\ell, u_r) = (0, 1)$ in case (b) recovers exactly u_B from Section 5.3, and taking $(u_\ell, u_r) = (1, 0)$ in case (a) gives the shock $s = 1/2$ analogous to u_A — now correctly identified as the entropy solution, since the roles of u_ℓ and u_r are reversed from Section 5 and the Lax condition (32) now holds.

8 Extensions

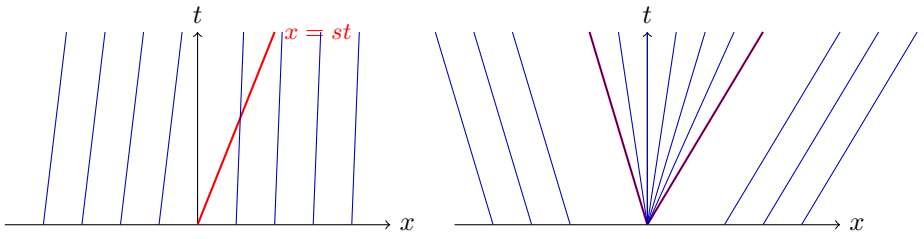
8.1 The Cole–Hopf transformation

This extension reintroduces viscosity and solves the viscous Burgers' equation (2) exactly, using the potential-flow structure noted in Remark 1.1.

Theorem 8.1 (Cole–Hopf transformation)**.** *Let u solve $u_t + u\,u_x = \nu\,u_{xx}$, and define $\varphi > 0$ by*

$$u = -2\nu\,\frac{\varphi_x}{\varphi}, \qquad \textit{i.e.} \qquad \varphi(x,t) = \exp\left(-\frac{1}{2\nu}\int_0^x u(y,t)\,dy\right). \tag{37}$$

Then φ solves the heat equation $\varphi_t = \nu\,\varphi_{xx}$.



(a) shock: $u_\ell > u_r$ (b) rarefaction: $u_\ell < u_r$

Figure 2: Characteristic pictures for the two cases of Theorem 7.1. (a) Characteristics from both sides converge into the shock curve $x = st$. (b) Characteristics fan out from the origin, filling the gap left by the diverging characteristics from u_ℓ and u_r.

Proof. Write $\psi = \ln \varphi$, so $u = -2\nu\,\psi_x$. Substituting into the viscous Burgers' equation and dividing by -2ν gives

$$\psi_{xt} - 2\nu\,\psi_x\psi_{xx} - \nu\,\psi_{xxx} = 0, \tag{38}$$

which is exactly $\partial_x\big[\psi_t - \nu\,\psi_{xx} - \nu\,(\psi_x)^2\big] = 0$. Hence $\psi_t - \nu\,\psi_{xx} - \nu\,(\psi_x)^2 = C(t)$ for some function C of t alone; since $u = -2\nu\,\psi_x$ is unchanged if ψ is shifted by a function of t, φ may be rescaled by $\exp\big(-\int_0^t C\big)$ without affecting u, so we may take $C \equiv 0$:

$$\psi_t = \nu\,\psi_{xx} + \nu\,(\psi_x)^2. \tag{39}$$

Since $\varphi = e^{\psi}$ gives $\varphi_t = \varphi\,\psi_t$ and $\varphi_{xx} = \varphi\big(\psi_{xx} + (\psi_x)^2\big)$, equation (39) is equivalent to $\varphi_t = \nu\,\varphi_{xx}$. □

8.2 Exact solution via the heat kernel

By Theorem 8.1, solving (2) reduces to solving the heat equation for φ with initial data

$$\varphi_0(x) = \exp\left(-\frac{1}{2\nu}\int_0^x u_0(y)\,dy\right), \tag{40}$$

which is given by convolution with the heat kernel,

$$\varphi(x,t) = \int_{\mathbb{R}} \frac{1}{\sqrt{4\pi\nu t}}\,\exp\left(-\frac{(x-y)^2}{4\nu t}\right)\varphi_0(y)\,dy. \tag{41}$$

Substituting (40)–(41) into $u = -2\nu\,\varphi_x/\varphi$ and differentiating under the integral gives the explicit *Cole–Hopf formula*

$$u(x,t) = \frac{\displaystyle\int_{\mathbb{R}} \frac{x-y}{t}\,\exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}{\displaystyle\int_{\mathbb{R}} \exp\left(-\frac{G(x,y,t)}{2\nu}\right)dy}, \qquad G(x,y,t) = \int_0^y u_0(s)\,ds + \frac{(x-y)^2}{2t}. \tag{42}$$

Corollary 8.2 (Global smoothness)**.** *For every bounded measurable u_0 and every $\nu > 0$, formula* (42) *defines the unique bounded solution of* (2)*, and this solution is C^∞ for all $t > 0$.*

Unlike the inviscid equation (Proposition 3.1), solutions of (2) stay smooth for all time, for any $\nu > 0$ however small: advection and viscous diffusion together do not by themselves produce singularities (Remark **??**).

8.3 The vanishing viscosity limit

The Cole–Hopf solution also explains why the Lax entropy condition of Section 6 is the physically correct selection principle, rather than an arbitrary one.

Theorem 8.3 (Vanishing viscosity limit; Hopf, 1950)**.** *Let $u_0 \in L^\infty(\mathbb{R})$, let u^ν denote the solution of* (2) *with data u_0 given by* (42)*, and let u denote the entropy solution of* (3) *with the same data (Theorem 7.1, extended to general u_0 by the same selection principle). Then $u^\nu \to u$ pointwise almost everywhere, and in L^1_{loc}, as $\nu \to 0^+$ [3].*

The proof (a Laplace-method asymptotic analysis of (42) as $\nu \to 0$, showing the integral concentrates near the y minimising $G(x,y,t)$) is not reproduced here. Its significance is conceptual: the entropy condition of Definition 6.1 is not an ad hoc rule imposed to force uniqueness, but exactly the criterion selected by the physically natural limit of vanishing viscosity — the same limit by which the inviscid equation (3) was justified as an idealisation of (2) in Section 1.

8.4 General convex flux

The theory of Sections 4–7 was developed for the specific flux $f(u) = u^2/2$, but, as noted in Section 1.2, none of it relies on this choice beyond convexity of f. This extension records how the main results generalise to an arbitrary $f \in C^2(\mathbb{R})$ with $f'' > 0$ everywhere.

Proposition 8.4 (General Rankine–Hugoniot and Lax conditions)**.** *For a weak solution of $u_t + f(u)_x = 0$ with a jump from u_ℓ to u_r along $x = s(t)$, the Rankine–Hugoniot condition reads*

$$s'(t) = \frac{f(u_\ell) - f(u_r)}{u_\ell - u_r}, \tag{43}$$

and the Lax entropy condition reads $f'(u_\ell) > s'(t) > f'(u_r)$. Since f is convex, f' is strictly increasing, so this is equivalent to $u_\ell > u_r$, exactly as in the Burgers' case (32)*.*

Proof sketch. The proofs of Theorem 4.3 and Proposition 6.3 used only integration by parts and the definition of the weak formulation, independent of the specific form of f, so they carry over verbatim. □

Proposition 8.5 (General Riemann solution)**.** *For f strictly convex, the entropy solution of $u_t + f(u)_x = 0$ with data* (33) *is:*

(a) *If $u_\ell > u_r$: the shock* (35) *with speed s given by* (43)*.*

(b) *If $u_\ell \le u_r$: the rarefaction fan*

$$u(x,t) = \begin{cases} u_\ell, & x \le f'(u_\ell)\,t, \\ (f')^{-1}(x/t), & f'(u_\ell)\,t < x < f'(u_r)\,t, \\ u_r, & x \ge f'(u_r)\,t, \end{cases} \tag{44}$$

where $(f')^{-1}$ is well defined since f' is strictly increasing.

Proof sketch. The self-similar ansatz $u(x,t) = g(x/t)$ substituted into $u_t + f(u)_x = 0$ gives $g'(\xi)\big(f'(g(\xi)) - \xi\big) = 0$ (the general-flux analogue of the computation in Section 7.1), so either g is constant or $f'(g(\xi)) = \xi$, i.e. $g(\xi) = (f')^{-1}(\xi)$. Taking $\xi = f'(u_\ell)$ and $\xi = f'(u_r)$ as the two matching boundaries recovers (44); continuity at both boundaries and vanishing of the corresponding boundary terms in the weak formulation follow exactly as in the proof of Theorem 7.1(b). Part (a) follows exactly as in Theorem 7.1(a), using Proposition 8.4 in place of (32). □

Remark 8.6. Taking $f(u) = u^2/2$ gives $f'(u) = u$ and $(f')^{-1}(\xi) = \xi$, recovering Theorem 7.1 exactly. The role of convexity is essential: it is what guarantees f' is strictly increasing, which is in turn what makes the Lax condition equivalent to the simple inequality $u_\ell > u_r$, and what makes $(f')^{-1}$ well defined in (44). For non-convex f, both the entropy condition and the Riemann solution are more intricate (they generally involve the convex hull of f between u_ℓ and u_r), and are not treated here.

8.5 A numerical scheme: Godunov's method

This extension outlines a numerical scheme for the conservation law $u_t + f(u)_x = 0$ that uses the exact Riemann solution of Theorem 7.1 as its building block, and states its convergence to the entropy solution.

8.5.1 Finite volume discretisation

Divide $\mathbb{R}$ into cells $[x_{j-1/2}, x_{j+1/2}]$ of width Δx, with cell averages

$$U_j^n \approx \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t^n)\, dx, \tag{45}$$

at times $t^n = n\Delta t$. Integrating $u_t + f(u)_x = 0$ over a cell and over $[t^n, t^{n+1}]$ gives the exact identity

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{\Delta x}\left(\widehat{f}_{j+1/2} - \widehat{f}_{j-1/2}\right), \tag{46}$$

where $\widehat{f}_{j+1/2}$ is the (time-averaged) flux through the interface $x_{j+1/2}$, which must be approximated by a *numerical flux function* $\widehat{f}_{j+1/2} = \widehat{f}(U_j^n, U_{j+1}^n)$.

8.5.2 The Godunov flux

Godunov's method defines $\widehat{f}$ by solving, at each interface, the Riemann problem with left and right states U_j^n and U_{j+1}^n *exactly*, using Theorem 7.1, and evaluating the resulting self-similar solution along $x/t = 0$:

$$\widehat{f}_{j+1/2} = f\big(u^R(0;\, U_j^n, U_{j+1}^n)\big), \tag{47}$$

where $u^R(\xi; u_\ell, u_r)$ denotes the value at self-similar coordinate $\xi = x/t$ of the entropy solution from Theorem 7.1 with data (u_ℓ, u_r). Concretely, by (35)–(36),

$$u^R(0; u_\ell, u_r) = \begin{cases} u_\ell, & u_\ell > u_r,\ s = \frac{u_\ell + u_r}{2} > 0, \\ u_r, & u_\ell > u_r,\ s < 0, \\ u_\ell, & u_\ell \le u_r,\ u_\ell \ge 0, \\ u_r, & u_\ell \le u_r,\ u_r \le 0, \\ 0, & u_\ell < 0 < u_r, \end{cases} \tag{48}$$

the last case corresponding to $\xi = 0$ lying inside the rarefaction fan of (36), i.e. $u^R(0) = (f')^{-1}(0) = 0$ for Burgers' flux.

8.5.3 Stability and convergence

Proposition 8.7 (CFL condition)**.** *The Godunov scheme is stable provided the time step satisfies*

$$\Delta t \le \frac{\Delta x}{\max_j |f'(U_j^n)|}. \tag{49}$$

Condition (49) (the Courant–Friedrichs–Lewy condition) ensures that information does not propagate across more than one cell per time step, consistent with the finite propagation speed of characteristics established in Section 2.

Theorem 8.8 (Convergence; see [2] and standard finite volume references)**.** *Under* (49)*, the Godunov approximations converge, as $\Delta x, \Delta t \to 0$, to the unique entropy solution of Theorem 6.4.*

Remark 8.9. This gives a second, independent route to the solutions constructed by hand in Sections 5–7: an implementation of (47) applied to the Riemann data (27) should reproduce the shock speed $s = 1/2$ and the linear profile $u = x/t$ inside the rarefaction fan, providing a computational check on the exact formulas derived analytically above.

8.5.4 Numerical results

The scheme of Section 8.5.1–8.5.2 was implemented with $N = 400$ cells on $x \in [-2, 2]$, using the CFL number 0.45 in (49), and run to $t = 1$ on both Riemann problems considered earlier: the rarefaction data $(u_\ell, u_r) = (0, 1)$ of Section 5, and the shock data $(u_\ell, u_r) = (1, 0)$. Figure 3 compares the numerical solution against the exact entropy solution of Theorem 7.1.

The measured L^1 errors between the numerical and exact solutions are approximately 4.5×10^{-3} (rarefaction case) and 1.2×10^{-3} (shock case), consistent with the first-order convergence rate of Theorem 8.8; the shock case converges faster per cell because the error is concentrated in the single cell straddling the discontinuity, whereas the rarefaction error accumulates gradually across the two fan corners.

8.6 Interaction of two shocks

This extension studies what happens when two shocks, moving at different speeds, collide. Remarkably, the answer follows directly from Theorem 7.1 without any new machinery.

8.6.1 Setup

Consider three constant states $u_1 > u_2 > u_3$ and initial data

$$u_0(x) = \begin{cases} u_1, & x < a, \\ u_2, & a < x < b, \\ u_3, & x > b, \end{cases} \qquad a < b. \tag{50}$$

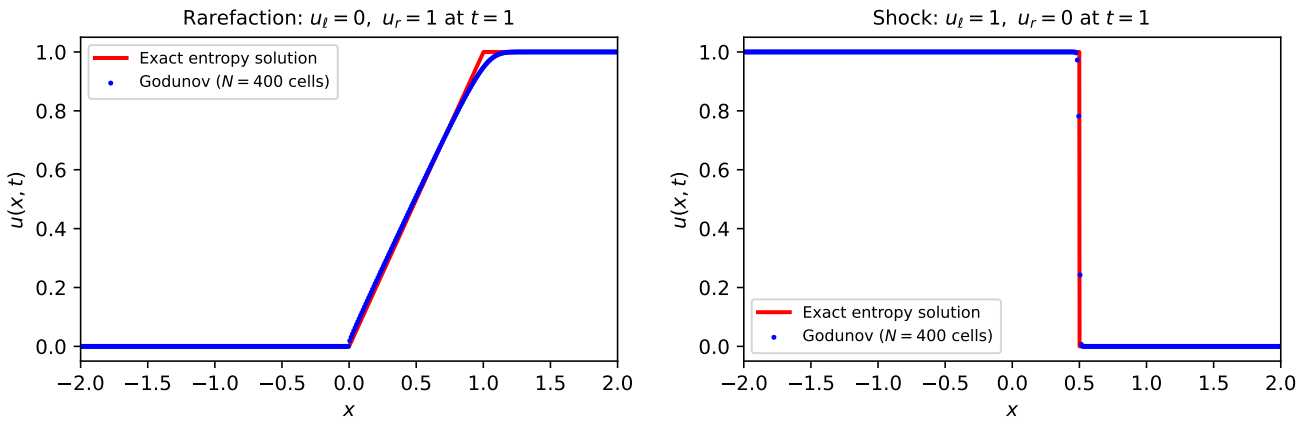


Figure 3: Godunov scheme (blue dots, $N = 400$ cells) against the exact entropy solution (red) of Theorem 7.1, at $t = 1$. Left: rarefaction data $(u_\ell, u_r) = (0, 1)$; the scheme reproduces the linear fan $u = x/t$, with mild smearing at the two corners typical of a first-order scheme. Right: shock data $(u_\ell, u_r) = (1, 0)$; the discontinuity is captured sharply at $x = st = 0.5$, exactly the Rankine–Hugoniot speed from (26), with no spurious oscillation.

Since $u_1 > u_2$ and $u_2 > u_3$, both jumps individually satisfy the Lax condition (32), so for small $t > 0$ the entropy solution consists of two shocks, moving at the Rankine–Hugoniot speeds

$$s_{12} = \frac{u_1 + u_2}{2}, \qquad s_{23} = \frac{u_2 + u_3}{2}, \tag{51}$$

with positions $x_L(t) = a + s_{12}\, t$ and $x_R(t) = b + s_{23}\, t$, exactly as in Theorem 7.1(a) applied locally at each jump.

8.6.2 Collision time

Proposition 8.10 (Shock collision)**.** *Since $u_1 > u_2 > u_3$, $s_{12} > s_{23}$, so the two shocks collide at the finite time*

$$t^\times = \frac{b - a}{s_{12} - s_{23}} = \frac{2(b - a)}{u_1 - u_3}, \tag{52}$$

at the point $x^\times = a + s_{12}\, t^\times$.

Proof. $s_{12} - s_{23} = \frac{1}{2}(u_1 + u_2) - \frac{1}{2}(u_2 + u_3) = \frac{1}{2}(u_1 - u_3) > 0$, so $x_L(t)$ and $x_R(t)$ are both affine in t with x_L increasing faster than x_R and $x_L(0) = a < b = x_R(0)$; they therefore meet at a unique finite time, found by solving $x_L(t^\times) = x_R(t^\times)$, which gives (52). □

8.6.3 The solution after collision

Theorem 8.11 (Merged shock)**.** *For $t \geq t^\times$, the entropy solution of* (50) *is the single shock*

$$u(x,t) = \begin{cases} u_1, & x < x^\times + s_{13}(t - t^\times), \\ u_3, & x > x^\times + s_{13}(t - t^\times), \end{cases} \qquad s_{13} = \frac{u_1 + u_3}{2}. \tag{53}$$

Proof. At $t = t^\times$, the middle state u_2 occupies the shrinking interval $[x_L(t), x_R(t)]$, which by construction has width 0 exactly at $t^\times$; hence at $t = t^\times$ the solution is precisely the two-state Riemann data $u(x, t^\times) = u_1$ for $x < x^\times$, u_3 for $x > x^\times$, with no intermediate state remaining. Since $u_1 > u_3$, this is exactly the hypothesis of Theorem 7.1(a) (shifted to start at time $t^\times$ and position $x^\times$ rather than 0 and 0, which does not affect the argument, as the equation is autonomous in both x and t), whose unique entropy solution for $t \geq t^\times$ is exactly the shock (53). □

Remark 8.12. Theorem 8.11 illustrates a general principle: an entropy solution's dependence on only the two bounding Riemann states means any intermediate structure that shrinks to a point is simply forgotten. This is also why the Godunov scheme of Section 8.5 handles merging shocks with no special logic: each time step re-solves a fresh Riemann problem at every interface.

8.7 A damped Burgers' equation

This extension adds a linear damping term,

$$u_t + u\,u_x = -\beta\,u, \qquad \beta > 0, \tag{54}$$

and asks how damping affects the shock formation result of Proposition 3.1. Equation (54) arises, for instance, as a model for weakly nonlinear waves subject to linear dissipation.

8.7.1 Characteristics with damping

Along a curve $x(t)$ with $\dot{x}(t) = u(x(t),t)$, the chain rule together with (54) gives

$$\frac{d}{dt}\,u(x(t),t) = u_t + u\,u_x = -\beta\,u(x(t),t), \tag{55}$$

so u no longer stays constant along characteristics, but decays exponentially:

$$u(x(t),t) = u_0(x_0)\,e^{-\beta t}. \tag{56}$$

Substituting (56) into $\dot{x}(t) = u(x(t),t)$ and integrating gives the characteristic curves explicitly:

$$x(t) = x_0 + u_0(x_0)\,\frac{1-e^{-\beta t}}{\beta}. \tag{57}$$

Unlike the undamped case, these are no longer straight lines, but they remain explicit, and the implicit solution formula generalises exactly as in Theorem 2.2: $u(x,t) = u_0(x_0)e^{-\beta t}$ where x and x_0 are related by (57), valid wherever

$$\Phi_t^{\beta}(x_0) := x_0 + u_0(x_0)\,\frac{1-e^{-\beta t}}{\beta} \tag{58}$$

is invertible.

8.7.2 A threshold for shock formation

Theorem 8.13 (Damping can prevent shock formation)**.** *Let $m = \min_{x_0} u_0'(x_0)$. If $m \geq -\beta$, the classical solution of (54) exists for all $t \geq 0$. If $m < -\beta$, a shock forms at the finite time*

$$T_\beta^* = -\frac{1}{\beta}\ln\!\left(1 + \frac{\beta}{m}\right). \tag{59}$$

Proof. As in the proof of Proposition 3.1, breakdown occurs at the first time some $(\Phi_t^{\beta})'(x_0) = 1 + u_0'(x_0)\,\frac{1-e^{-\beta t}}{\beta}$ vanishes. Writing $\tau(t) = (1-e^{-\beta t})/\beta$, this requires $\tau(t) = -1/u_0'(x_0)$ for some x_0, which (since τ is increasing) is hardest to achieve for the most negative value of u_0', namely x_0 attaining m. Since τ increases monotonically from $\tau(0) = 0$ towards $\tau(\infty) = 1/\beta$ without ever reaching it, a finite solution t to $\tau(t) = -1/m$ exists if and only if $-1/m < 1/\beta$, i.e. $m < -\beta$; solving $\tau(T_\beta^*) = -1/m$ for T_β^* gives (59). If instead $m \geq -\beta$, then $-1/m \geq 1/\beta \geq \tau(t)$ for every finite t (interpreting $-1/m = \infty$ if $m = 0$), so $(\Phi_t^{\beta})'(x_0) > 0$ for every x_0 and every $t \geq 0$, and the classical solution persists for all time by the same argument as Theorem 2.2. □

Remark 8.14. Setting $\beta = 0$ recovers Proposition 3.1 exactly, as can be checked with L'Hôpital's rule on $-\frac{1}{\beta}\ln(1+\beta/m)$ as $\beta \to 0$. More strikingly, the theorem shows that for *any* initial data, however steep, taking $\beta \geq -m$ prevents shock formation entirely: sufficiently strong linear damping smooths out the steepening mechanism of Section 3 before characteristics can converge.

8.8 Long-time behaviour: the N-wave

This final extension describes the universal shape that entropy solutions with compactly supported initial data approach as $t \to \infty$, regardless of the detailed shape of u_0.

8.8.1 An exact self-similar solution

Proposition 8.15 (The N-wave)**.** *Fix constants $P, Q > 0$. The function*

$$N(x,t) = \begin{cases} 0, & x \le -\sqrt{2Qt}, \\ x/t, & -\sqrt{2Qt} < x < \sqrt{2Pt}, \\ 0, & x \ge \sqrt{2Pt}, \end{cases} \qquad t > 0, \tag{60}$$

is an entropy solution of $u_t + u\,u_x = 0$ on $\mathbb{R} \times (0,\infty)$.

Proof. On the open middle interval, $N(x,t) = x/t$ is exactly the self-similar rarefaction profile of Section 7.1, which solves the PDE classically there; the outer regions are trivially constant solutions. At the right edge $x_+(t) = \sqrt{2Pt}$, the one-sided limits are $u_\ell = x_+(t)/t = \sqrt{2P/t}$ and $u_r = 0$; since $\dot{x}_+(t) = \sqrt{P/(2t)} = \frac{1}{2}\big(\sqrt{2P/t} + 0\big)$, this is exactly the Rankine–Hugoniot speed (26), and $u_\ell > u_r$ so the Lax condition (32) holds. The same check at the left edge $x_-(t) = -\sqrt{2Qt}$, where $u_\ell = 0$ and $u_r = -\sqrt{2Q/t}$, gives $\dot{x}_-(t) = -\sqrt{Q/(2t)} = \frac{1}{2}\big(0 + (-\sqrt{2Q/t})\big)$, again matching (26), with $u_\ell = 0 > -\sqrt{2Q/t} = u_r$ satisfying the Lax condition. Hence both jumps are entropy-admissible, and N is an entropy solution by Theorem 4.3 and Definition 6.1. □

The profile (60) is called an *N-wave*: it rises linearly through the origin and is bounded by a shock on each side, so that its graph traces the shape of the letter N (reversed), decaying in amplitude like $t^{-1/2}$ and spreading in width like $t^{1/2}$ as $t \to \infty$. It is an exact solution for every $t > 0$, though its formal limit as $t \to 0^+$ is not an ordinary function of the type considered in Section 4: both edges collapse to the origin, consistent with $N(\cdot,t)$ representing, in the limit, a state concentrated entirely at a point.

8.8.2 Convergence of general solutions to the N-wave

Theorem 8.16 (Asymptotic N-wave decay; Lax, 1957 [4])**.** *Let $u_0 \in L^1(\mathbb{R})$ have compact support, and set*

$$P = \int_{\mathbb{R}} \max(u_0(x), 0)\,dx, \qquad Q = \int_{\mathbb{R}} \max(-u_0(x), 0)\,dx. \tag{61}$$

Then the entropy solution u of (6) *satisfies*

$$\|u(\cdot,t) - N(\cdot,t)\|_{L^1(\mathbb{R})} \to 0 \qquad \text{as } t \to \infty, \tag{62}$$

where N is the N-wave (60) *with the same constants P, Q.*

The proof is not reproduced here, but the mechanism is exactly the shock-merging phenomenon of Section 8.6: a compactly supported u_0 produces, for small t, a solution with possibly many shocks and rarefactions, but by Theorem 8.11 every pair of shocks that meet coalesces into one, so only finitely many discontinuities survive as $t \to \infty$. Since the total signed area $\int u(x,t)\,dx = P - Q$ is conserved for all t (integrate (22) over all x), the surviving structure is forced into the unique self-similar shape consistent with these boundary areas — exactly (60).

Remark 8.17. This result explains the term "N-wave" as used in acoustics for the pressure signature of a sonic boom: an initially localised disturbance, regardless of its detailed shape, is predicted by the same mechanism to decay into the universal N-shaped profile of Proposition 8.15 once nonlinear and dissipative effects have had enough time to act, with amplitude decaying as $t^{-1/2}$, exactly as observed for sonic booms far from their source.

9 Conclusion

This paper has given a complete account of Burgers' equation, from its origin as a simplified model of nonlinear advection and viscous diffusion (Section 1) to the theory required to solve it once classical solutions cease to exist. The method of characteristics (Section 2) gives an explicit solution formula that is exact but necessarily temporary: Section 3 showed that decreasing initial data produces a genuine, finite-time loss of smoothness, with an exact formula for the time and location of the first singularity. Making sense of the equation beyond this point required weakening the notion of solution (Section 4), which in turn revealed a genuine failure of uniqueness (Section 5): the same initial data can evolve into more than one weak solution. The Lax entropy condition (Section 6) was introduced precisely to resolve this, and Section 7 used it to give the complete, explicit entropy solution of the Riemann problem in both possible cases.

The extensions in Section 8 showed that this core theory extends well beyond the specific flux $u^2/2$. The Cole–Hopf transformation justified the entropy condition itself, as the selection principle recovered in the vanishing-viscosity

limit rather than an arbitrary rule, and showed that viscosity alone, without the vortex-stretching mechanism absent from this one-dimensional model, is enough to guarantee global smoothness — supporting Remark **??**. The general convex flux extension confirmed none of the core results are special to Burgers' particular flux. The Godunov scheme, built directly on the exact Riemann solver derived here, reproduced the analytic shock speed and rarefaction profile numerically to within first-order error. The shock interaction result showed multiple shocks merge via the same Riemann-problem mechanism used throughout, the damped equation showed this mechanism can be delayed or prevented by sufficiently strong linear dissipation, and the N-wave result showed that the long-time behaviour of any compactly supported solution is governed by a single universal profile, determined only by the total positive and negative area of the initial data.

Taken together, these results illustrate why Burgers' equation, despite its simplicity, remains a standard reference point in the theory of nonlinear partial differential equations: it is simple enough to solve exactly and completely, in every regime considered here, while already exhibiting the central phenomena — shock formation, non-uniqueness of weak solutions, the necessity of an entropy condition, and the regularising effect of viscosity — that make the general theory of nonlinear conservation laws, and by extension the far harder open questions surrounding the Navier–Stokes and Euler equations, so difficult to resolve.

References

[1] J. M. Burgers. A mathematical model illustrating the theory of turbulence. *Advances in Applied Mechanics*, 1:171–199, 1948.

[2] Lawrence C. Evans. *Partial Differential Equations*. American Mathematical Society, 2nd edition, 2010.

[3] Eberhard Hopf. The partial differential equation $u_t + uu_x = \mu u_{xx}$. *Communications on Pure and Applied Mathematics*, 3(3):201–230, 1950.

[4] Peter D. Lax. Hyperbolic systems of conservation laws ii. *Communications on Pure and Applied Mathematics*, 10(4):537–566, 1957.